

Public Economics and Development

Week 4: Classic Optimal Taxation Theories

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Roadmap for today

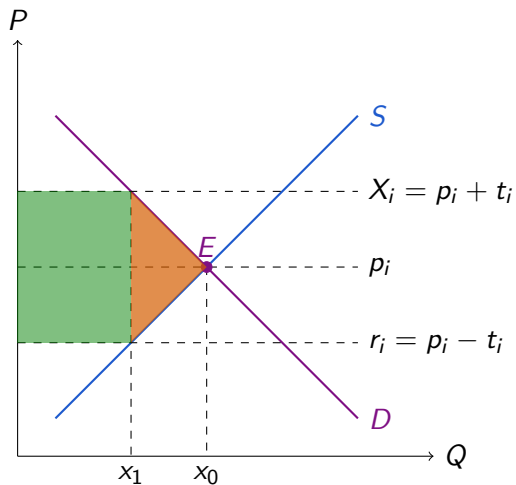
1. Basic tools of public economics
2. Optimal Consumption Taxation
 - 2.1 Consumer choice and deadweight loss
 - 2.2 Ramsey taxation rules
3. Optimal Income Taxation
 - 3.1 Linear taxation
 - 3.2 Nonlinear taxation
4. Conclusion

Public economics as study of taxation

Quantifying the gains and costs of taxation

- All taxes are based on observable indicators - labor earnings, transactions etc
- People respond to taxes by adjusting their behaviors
 - ▶ Income tax $\uparrow \Rightarrow$ Opportunity cost of labor $\uparrow \Rightarrow$ Labor $\downarrow$, Leisure $\uparrow$
 - ▶ Consumption tax $\uparrow \Rightarrow$ Consumer price $\uparrow \Rightarrow$ Transaction $\downarrow$
- This leads to efficiency costs (deadweight loss): lower labor and lost transactions
 - ▶ Lost production and income due to less labor
 - ▶ Transactions that would have occurred pre-tax no longer exists
- How to capture this cost, and is this cost bearable? (more on this in normative analysis)

Graphical description of taxes in supply-demand framework



Pre-tax

- At equilibrium E : price p_i , quantity x_0
- Consumer surplus: Above p_i below D
- Producer surplus: Above S , below p_i

Post-tax setup: Government levies tax t_i

- Consumer price: X_i , Producer price: r_i
- Transaction reduced from x_0 to x_1
- Both surpluses fall
- Green: Tax revenues (to government)
- Red: Deadweight loss (lost transactions)

Public economics as study of distribution

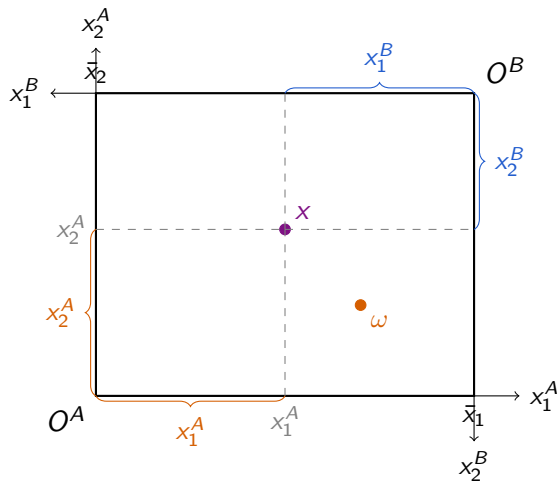
Normative analysis: What is a desirable allocation of resources and how to achieve it?

- Allocation: How much of each good does each individual have after transaction?
 - ▶ Assume n individuals and k goods exist (for simplicity, $n = k = 2$)
 - ▶ $x^i \equiv (x_1^i, x_2^i, \dots, x_k^i)$ is allocation of goods $x_1, \dots, x_k$ for consumer i
 - ▶ $x \equiv (x^1, x^2, \dots, x^n)$ is an allocation across n individuals
- Each individual derives utility from their consumption: $u^i = u^i(x_1^i, x_2^i, \dots, x_k^i)$
 - ▶ $u_i' > 0$: Marginal utility is positive, $u_i'' < 0$ Diminishing marginal utility

Assessing allocations with Pareto efficiency framework

- Pareto improvement: Reallocation of resources from allocation x to y that
 - ▶ Leaves everyone at least equal (no one is worse off)
 - ▶ At least one person is strictly better off
- Pareto efficient: An allocation where no reallocation satisfies above conditions

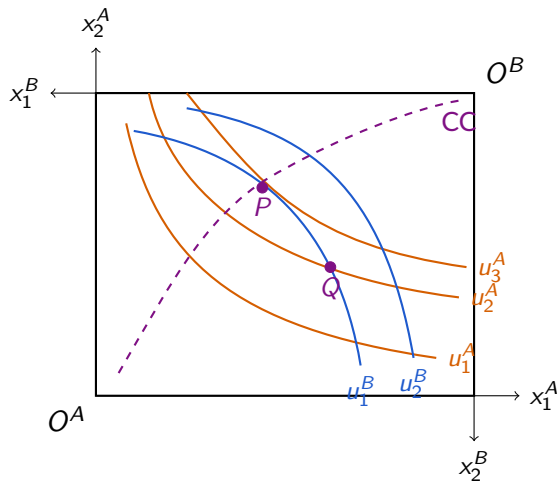
Graphical analysis of exchange economy with Edgeworth box



Edgeworth box

- Individual A,B and goods 1,2
- Total endowments: $\bar{x}_1$ and $\bar{x}_2$
- Every point inside the box is a feasible allocation
- O^i : origin for $i \in \{A, B\}$
- Point ω : initial endowment (pre-trade allocation)
- x is one possible trade
- Is this desirable?

Pareto efficiency and contract curve



Pareto efficient allocation

- u_1^A, u_2^A : A's indifference curves
- u_1^B, u_2^B : B's indifference curves
- Q: Current allocation
 - ▶ u_A rises with $x_1^A \downarrow, x_2^A \uparrow$
 - ▶ u_B rises with $x_2^A \uparrow, x_1^A \downarrow$
 - ▶ Q is thus not Pareto efficient
- P: Pareto efficient
 - ▶ Any departure: $u^A \downarrow$ or $u^B \downarrow$
 - ▶ Tangent indifference curves
- CC: contract curve — locus of all Pareto efficient allocations

Pareto efficiency in mathematics

Conditions for Pareto efficient allocation

- Marginal rate of substitution between good 1 and 2 ($MRS_{1,2} = \frac{MU_1}{MU_2} = \frac{u'_1}{u'_2}$)
 - ▶ Rate at which consumer is willing to trade two goods while keeping utility same
 - ▶ Subjective value of good 1 in terms of good 2 that keeps utility same
 - ▶ It is the slope of the indifference curves
- Pareto efficient allocation in exchange economy: $MRS_{1,2}^A = MRS_{1,2}^B$
 - ▶ Indifference curves of both A and B are tangent to each other
- Intuitively, suppose $MRS_{1,2}^A = 1$ and $MRS_{1,2}^B = 2$
 - ▶ A is willing to trade 1 unit of good 1 with 1 unit of good 2
 - ▶ B is willing to trade 1 unit of good 1 with 2 units of good 2
 - ▶ Suppose A gives up 1 unit of good 1 and gets 2 units of good 2
 - ▶ $u^A \uparrow$ by an amount equal to one extra unit of good 1, while u^B stays the same
 - ▶ Thus, a pareto improvement is possible $\rightarrow$ pre-trade allocation is not pareto efficient

Two welfare theorems

Two Welfare Theorems

- First Welfare Theorem: A competitive equilibrium is Pareto efficient
- Second Welfare Theorem: Any Pareto efficient allocation can be achieved as a competitive equilibrium, following an appropriate redistribution of endowments

Takeaways

- Markets can achieve Pareto efficient allocation without intervention (1st thm)
 - ▶ Competitive prices lead each consumer to set $MRS_{12}^i = p_1/p_2$ for all i
- However, Pareto efficient allocations may not be desirable in terms of equality
 - ▶ Any allocation on the contract curve is Pareto efficient
- Taxes can redistribute toward a socially preferred allocation (2nd)
- That comes with welfare costs (deadweight loss quantified earlier)

Formalizing the redistribution questions

Social Welfare Function

- A function reflecting how utilities of its members affect the society as a whole
- Generic notation

$$W = F(u^A, u^B)$$

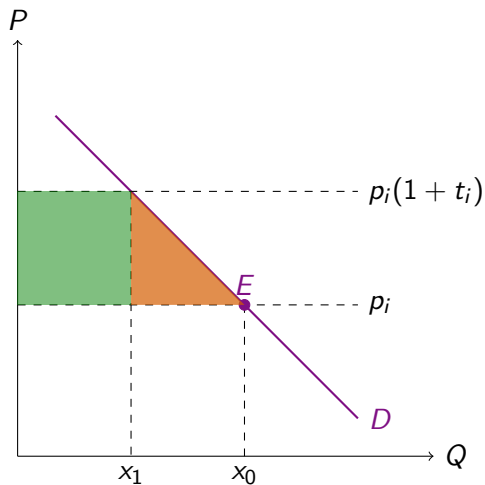
- Different functional forms and tax instruments lead to different conclusions
 - ▶ Utilitarian function: $u^A + u^B$ - Simple sum of utilities
 - ▶ Weighted sum of utilities: $\alpha u^A + (1 - \alpha)u^B$ - Can put different weights on A and B
 - ▶ Rawlsian function: $\min \{u^A, u^B\}$ - Maximizing the welfare of the poorest
- Government maximizes social welfare function subject to budget constraint, typically

$$\max_{t_i} W \text{ subject to (s.t.) } \sum_i t_i x_i \geq R$$

i.e. its (tax) revenues $\geq R$ (to be spent on public goods and redistribution)

Optimal Consumption Taxation

Calculating deadweight loss with one good



Tax rate of t_i

- $\Delta X = X_1 - X_0 (< 0)$, and $\Delta P = p_i t_i (> 0)$
- Calculating the triangle: $-\frac{1}{2} \Delta X \Delta P$

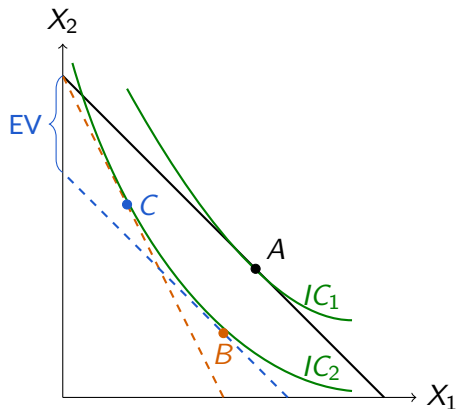
More general expression of deadweight loss

- Demand elasticity: $\epsilon_d = -\frac{\Delta X/X}{\Delta P/P} = -\frac{\Delta X}{\Delta P} \frac{P}{X}$
- From this, we can write deadweight loss as

$$DWL = \frac{1}{2} \underbrace{\left(\epsilon_d \Delta P \frac{X}{P} \right)}_{-\Delta X} \Delta P = \frac{1}{2} \epsilon_d P X t^2$$

- Larger DWL with higher tax and elasticity

Analyzing effects of price change with income and substitution effects



Suppose price of X_1 rises twofold

- Budget constraint tilts inward
- Consumption $A \rightarrow C$, Utility $IC_1 \rightarrow IC_2$

Question to quantify welfare changes

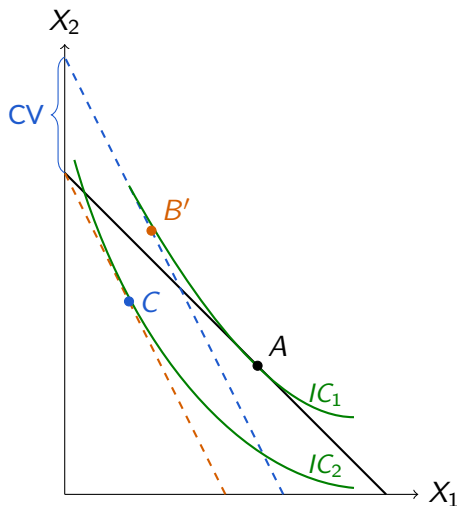
- What amount of change in income (based on old prices) lead to new IC?

Income and substitution effects

- Income: Changes in (real) income: $A \rightarrow B$
- Substitution: Change in relative price: $B \rightarrow C$

Note: Change in income (before price change) leading to new IC is *Equivalent variation* (EV)

Other way to analyze effect of price changes



Same situation, but different question

- How much income compensation (based on new prices) are required to return to old IC?

Income and substitution effects

- Substitution: x_2 relatively cheaper: $A \rightarrow B'$
- Income: (Real) income falls: $B' \rightarrow C$

The income compensation required to preserve old IC is *compensating variation* (CV)

- EV and CV usually differ ($\because$ different IC used)

Ad valorem tax vs lump sum tax

Advalorem tax

- A tax that is proportional to the transaction, such that

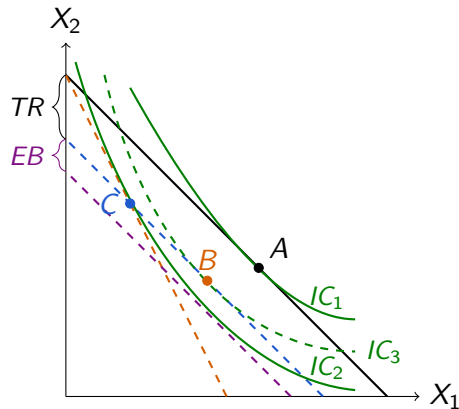
$$P_{\text{post-tax}} = P(1 + t)$$

- Budget constraint: $P_1X_1(1 + t_1) + P_2X_2 = I$
- Distortionary: Change relative price $\rightarrow$ Shift consumption to untaxed good
- Substitution and income effect present

Lump-sum taxation

- A tax where no change in behavior can affect the amount (same tax no matter what)
- Budget constraint: $P_1X_1 + P_2X_2 = I - T$
- No change in relative price, only income effect

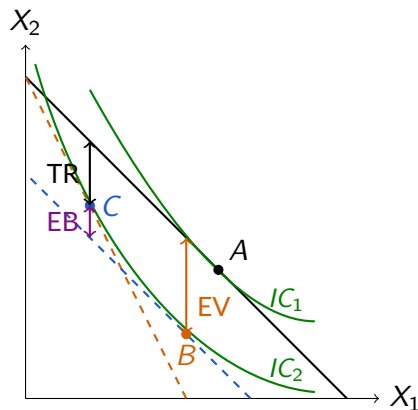
Excess burden: Hypothetical lump-sum vs consumption tax



Ad valorem and lump-sum with equal TR

- Lump-sum tax: $A \rightarrow B$, and $IC_1 \rightarrow IC_3$
- Ad valorem tax: $A \rightarrow C$, and $IC_1 \rightarrow IC_2$
- Assuming both are normal goods, $IC_2 < IC_3$
- EB : The difference in the utility
- This captures additional utility from distortionary taxes

Excess burden: Loss in utility not made up for in taxes



Consumption tax on X_1 (X_2 not taxed)

- Consume $A \rightarrow C$, Utility $IC_1 \rightarrow IC_2$
- EV : Welfare loss from tax
- TR : Tax revenue in terms of X_2 per unit of X_1
- EB is not made up for by anyone
- Thus, EB captures efficiency cost from distortionary taxation

Taxation in practice

Lump-sum tax must be independent of all observable behavior

- In reality, this is not the case
- Observable proxies – income, consumption etc – are subject to behavioral response
 - ▶ People underreport income/wealth, shift compensation to non-taxable forms
- Ideal lump-sum tax is based on innate ability, which is *private information*
 - ▶ No way to objectively verify: Some individuals can pretend to be of different type
- Thus, *all* taxes are distortionary → Optimal tax to minimize efficiency cost

Why tax at all if distortions exist?

- Distortions are measured relative to a perfectly competitive market
- In reality, markets failures exist → Taxation may improve welfare
 - ▶ Externalities, monopolies, public goods, information asymmetry
 - ▶ In developing countries, markets may not even exist → Government as sole provider

Optimal taxation with multiple goods: Inverse elasticity rule

Setup

- Assume two goods – X_1 and X_2 – are being taxed
 - Both goods are independent of each other (neither complement nor substitute)
- Optimal taxes minimize deadweight loss while obtaining tax revenue R

$$\text{Objective: } \min_{t_1, t_2} \frac{1}{2} [P_1 X_1 \epsilon_1 t_1^2 + P_2 X_2 \epsilon_2 t_2^2]$$

$$\text{Constraint: } t_1 P_1 X_1 + t_2 P_2 X_2 \geq R$$

- Solve : $\mathcal{L} = \frac{1}{2} [P_1 X_1 \epsilon_1 t_1^2 + P_2 X_2 \epsilon_2 t_2^2] + \lambda [t_1 P_1 X_1 + t_2 P_2 X_2 - R]$
 - F.O.C.[t_i]: $P_i X_i \epsilon_i t_i + P_i X_i \lambda = 0$ for $i \in \{1, 2\}$
- The inverse elasticity rule

$$\frac{t_1}{t_2} = \frac{\epsilon_2}{\epsilon_1} = \frac{1/\epsilon_1}{1/\epsilon_2}$$

Optimal taxation with multiple goods: Inverse elasticity rule

Implications of the inverse elasticity rule

- Using $\epsilon_i = -\frac{\Delta X_i}{X_i} \frac{P_i}{\Delta P_i}$ and $\Delta P_i = P_i t_i$, we can write

$$\frac{t_1}{t_2} = \frac{\epsilon_2}{\epsilon_1} \iff \frac{t_1}{t_2} = \frac{-\frac{\Delta X_2}{X_2} \frac{1}{t_2}}{-\frac{\Delta X_1}{X_1} \frac{1}{t_1}} \iff \frac{\Delta X_1}{X_1} = \frac{\Delta X_2}{X_2}$$

- Tax rate should be higher for goods for those with lower elasticity
- Tax rate should be set to equalize percent change in quantities demanded

Equity implications

- Necessities tend to have lower elasticity, luxuries higher
- Could result in tax burden going to low-income persons

Generalized version: Ramsey taxation rule (Ramsey (1927))

Setup

- Now we assume that demand for X_1 and X_2 can be interdependent
 - i.e. Tax of X_1 affects demand for X_1 and X_2

Define Cross-elasticity of good 1 and 2 as

$$\epsilon_{1,2} = \frac{\partial X_1 / X_1}{\partial P_2 / P_2} = \frac{\partial X_1}{X_1} \frac{P_2}{\partial P_2}$$

- Change in demand for good 1, $X_1(P_1, P_2)$ can be broken into (total differentiation)

$$\Delta X_1 = \frac{\partial X_1}{\partial P_1} \Delta P_1 + \frac{\partial X_1}{\partial P_2} \Delta P_2 \iff \epsilon_{1,1} \frac{X_1}{P_1} \Delta P_1 + \epsilon_{1,2} \frac{X_1}{P_2} \Delta P_2$$

- Across two goods, total deadweight loss is now

$$DWL = \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^2 \left[\left(\epsilon_{i,j} \frac{X_i}{P_j} \Delta P_j \right) \Delta P_i \right] = \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^2 \epsilon_{i,j} P_i X_i t_i t_j$$

Generalized version: Ramsey taxation rule (Ramsey (1927))

Solving for the optimal tax

- Lagrangian takes the following form

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^2 \epsilon_{i,j} P_i X_i t_i t_j + \mu [t_1 P_1 X_1 + t_2 P_2 X_2 - R]$$

- First order conditions on tax rates

$$[t_i] : \frac{1}{2} [2\epsilon_{i,i} P_i X_i t_i + \epsilon_{i,j} P_i X_i t_j + \epsilon_{j,i} P_j X_j t_j] + \mu P_i X_i = 0$$

- We rely on Slutsky symmetry: Cross-price substitution effect of good i on good j = Cross-price substitution effect of good j on good i

$$\frac{\partial X_i}{\partial P_j} = \frac{\partial X_j}{\partial P_i} \iff \epsilon_{i,j} \frac{X_i}{P_j} = \epsilon_{j,i} \frac{X_j}{P_i} \iff \epsilon_{i,j} X_i P_i = \epsilon_{j,i} X_j P_j$$

Generalized version: Ramsey taxation rule (Ramsey (1927))

Solving for the optimal tax

- The first order conditions become

$$\epsilon_{i,i} P_i X_i t_i + \epsilon_{i,j} P_i X_i t_j = -\mu P_i X_i \iff \sum_{j=1}^2 \epsilon_{i,j} t_j = -\mu$$

Intuition

- Elasticity weighted sum of tax rates should equal a constant
- That means elastic goods are taxed low, inelastic goods high
- Low-income persons relying heavily on necessities worse off?
- Solve with many-person framework: Higher welfare weights to low-income persons

Optimal Income Taxation

Design of optimal income tax: How to balance equity and efficiency

Equity motivation

- Individuals differ in earning capacity w – ability, family background, human capital
- These differences are largely outside individual control
- If $u''(c) < 0$, transferring \$1 from rich to poor raises total welfare

$$u'(c_L) > u'(c_H) \iff \text{marginal utility higher for the poor}$$

- Redistribution raises social welfare – but trade-off with efficiency cost exists

The fundamental tension: Balancing efficiency and equity

- We want to tax according to ability w (vertical equity)
- But w is *unobservable*, only income $z = w/l$ is observable
- Taxing z penalizes effort alongside ability – efficiency cost of redistribution
- Yet failing to redistribute leaves inequality from factors outside individual control

Equity consideration in taxes: Edgeworth Model

Individuals utilities

- Individuals have same utility function $u(\cdot)$ that differs only in income
- Pre-tax income z_i is exogenously given, with tax being T_i (lump-sum)
- All remaining income is consumed: $c_i = z_i - T_i$

Government's decision on taxes

- Government maximizes sum of individual utilities
- Subject to government budget constraint: Tax revenue at least greater than R
- Lagrangian setup is as follows

$$\mathcal{L} = \max_{T_i} \sum_{i=1}^n u(c_i) + \lambda \left[\sum_{i=1}^n T_i - R \right] \quad \text{where } c_i = z_i - T_i$$

- ▶ Replace c_i with $z_i - T_i$ and FOC w.r.t. T_i

$$-u'(z_i - T_i) + \lambda = 0 \iff u'(c_i) = \lambda \text{ for all } i$$

Equity consideration in taxes: Edgeworth Model

Takeaways from the conclusion

$$u'(c_i) = \lambda \text{ for all } i$$

- Equal marginal utility of consumption for all
- Since all have same preferences, taxes should be levied to equalize consumption
- This can be seen as full insurance
- As consumption is equalized, no incentive to work

What modern optimal taxation achieves

- Balance between efficiency (minimize lost utilities) and equity (fairness)
- Equity measured in terms of consumption inequality
- Efficiency in terms of labor supply or productivity

Linear income tax with labor supply decisions: Household decisions

- Representative individual with wage w chooses labor supply l to maximize $u(c, l)$ with $u_c = \frac{\partial u}{\partial c} > 0$, $u_l = \frac{\partial u}{\partial l} < 0$, facing:

$$c = (1 - t)wl + a, \quad z = wl$$

where t is the flat tax rate and a is the lump-sum transfer (or tax if $a < 0$)

- FOCs for Lagrangian $\mathcal{L} = u(c, l) + \lambda[(1 - t)wl + a - c]$:

$$\text{w.r.t } [c] : \quad u_c - \lambda = 0$$

$$\text{w.r.t } [l] : \quad u_l + \lambda(1 - t)w = 0$$

$$\Rightarrow \quad -\frac{u_l}{u_c} = (1 - t)w$$

- Marginal disutility of labor in consumption units = marginal gain from work

► $t \uparrow \rightarrow (1 - t)w \downarrow \rightarrow -\frac{u_l}{u_c} > (1 - t)w \rightarrow l \downarrow$ to restore equality $\rightarrow z \downarrow$

Linear income tax with labor supply decisions: Government problem

- Earnings $z(1 - t) = w/l$ respond to net-of-tax rate $(1 - t)$ via labor supply decision.
- Government seeks to maximize revenue $R = t \cdot z(1 - t)$
- Define earnings elasticity with respect to net-of-tax rate:

$$e = \frac{\partial z}{\partial(1 - t)} \frac{(1 - t)}{z} > 0$$

- Differentiating w.r.t t , we get:

$$\frac{dR}{dt} = z + t \frac{\partial z}{\partial t} = z - t \frac{\partial z}{\partial(1 - t)} = z \left(1 - \frac{t}{1 - t} e \right)$$

- Setting $\frac{dR}{dt} = 0$ gives the revenue-maximizing rate: $t = \frac{1}{1+e}$
 - ▶ Any higher tax rates $\rightarrow$ Revenue falls $\rightarrow$ This is the laffer curve argument
 - ▶ Requires large e , but this argument is not realized as e is known to be low (0.1 to 0.5)

Linear income taxation: Optimal tax rate

With two types, raising t by dt has two effects:

- Let $W(c_H, c_L)$ be the social welfare function
- *Equity gain*: transfers resources from H to L , raising social welfare if $W_{c_L} > W_{c_H}$
- *Efficiency cost*: both types work less, shrinking the tax base by e

Optimal t equates these two forces:

$$\underbrace{\frac{t}{1-t} \cdot e}_{\text{efficiency cost}} = \underbrace{\frac{W_{c_L} - W_{c_H}}{W_{c_H}}}_{\text{equity gain}}$$

- Higher $e \Rightarrow$ lower t^* ; stronger redistributive preference $\Rightarrow$ higher t^*
- Any chance to tax high and low types differently? $\rightarrow$ Nonlinear taxation

Nonlinear income taxation: Setup

Why not assign person-specific taxes $T_H \neq T_L$?

- There are a share of type H (and $1 - a$ of L)
- Government observes income $z = wl$ but *not* ability w directly
- Type H can mimic type L by working less (but keep ability): $l'_H = \frac{z_L}{w_H} < l_L$
- Since $w_H > w_L$, mimicking costs H less effort – taxes for L types may unravel

Constraints of the government

- **Incentive compatibility (IC):**

$$u\left(c_H, \frac{z_H}{w_H}\right) \geq u\left(c_L, \frac{z_L}{w_H}\right) \quad u\left(c_L, \frac{z_L}{w_L}\right) \geq u\left(c_H, \frac{z_H}{w_L}\right)$$

- **Budget constraint:**

$$a \times \underbrace{(z_H - c_H)}_{\text{Tax on } H = T(z_H)} + (1 - a) \times \underbrace{(z_L - c_L)}_{\text{Tax on } L = T(z_L)} \geq E$$

Nonlinear income taxation: Which IC binds?

Not all ICs matter

- Two IC constraints — but only one binds at the optimum
- **Downward IC** (H mimicking L) is the binding one:
 - ▶ H finds mimicking L relatively cheap — high wage means less effort to earn z_L
 - ▶ L mimicking H is prohibitively costly — low wage requires enormous effort to earn z_H
- So L 's IC is slack at the optimum — set it aside and focus on H 's IC binding:

$$u\left(c_H, \frac{z_H}{w_H}\right) = u\left(c_L, \frac{z_L}{w_H}\right)$$

H must be just indifferent between own bundle and L 's bundle (with its own ability)

- Any more redistribution to L would cause H to mimic L — IC violated
- Better utility for H with higher C_H ?
 - ▶ Reduce C_H and increase C_L by slight amounts $\rightarrow$ Social welfare $\uparrow$ ($\because u'(c_H) < u'(c_L)$)
- The binding IC is the key constraint limiting redistribution

Nonlinear income taxation: Key results (Mirrlees, 1971)

No distortion at the top: $T'(z_H) = 0$

- If $T'(z_H) > 0$, reducing it raises H 's labor supply
- It does not affect L 's labor or consumption $\rightarrow$ Pareto Improvement
- Intuition: no one above H to redistribute to – distorting H is pure efficiency loss

Positive distortion at the bottom: $T'(z_L) > 0$

- Distorting L downward makes mimicking less attractive for H
- Positive marginal rate for L is the necessary cost of redistribution under hidden ability

Caveats in practice

- Ability is not fixed (human capital investments)
- Top of the distribution is unknown, thus hard to know where to apply zero rate
- Even richer models with dynamic taxation and endogenous skill formation

Conclusion: Takeaways and open questions

The Unified Logic of Optimal Taxation

All three frameworks solve the same problem:

- Raise revenue
- Achieving redistribution
- Minimize efficiency costs

They differ in *what the government can observe*

Framework	Government observes	Key result
Consumption tax	Goods, not utility	Inverse elasticity rule
Income tax	Income, not ability	Zero top marginal tax

As the government learns more about the taxpayer, the optimal tax system becomes more personalized – and more efficient

What the Theory Cannot Resolve

Open tensions worth keeping in mind:

- **The zero top rate:** clean in theory, but requires *knowing where the distribution ends*.
- **Correct social welfare function?:** Results are sensitive to the social welfare function.
- **Dynamic taxation:** Types may change in reality (human capital accumulation). Taxing high earners today may discourage skill acquisition
- **Informality:** In developing economies, a large share of workers are outside the tax net entirely – different theory required
- **Political economy:** Politicians may favor visible tax cuts over socially optimal design

Theory and practice diverge in many settings – especially in developing countries. Is the gap a failure of policymakers to learn from theory, or of theory to reflect the constraints policymakers actually face?